

A Pixel-to-Dimension Rule for Pooling Earth Embeddings

Isaac Corley, Caleb Robinson, Inbal Becker-Reshef, and Juan M. Lavista Ferres

Abstract—Geospatial foundation models (GFMs) increasingly publish pixel-level embedding products that practitioners reuse without access to the encoder. Downstream patch- and region-level tasks must aggregate these dense embeddings into a single vector, yet pooling is usually treated as a default preprocessing step. We show that the ratio N/D of pixels per region to embedding dimension predicts when higher-order pooling is reliable: fixed chips with $N \gg D$ benefit from distributional statistics, while variable-size parcels with $N \approx D$ expose covariance pooling as an unstable estimator. Across EuroSAT and PASTIS embeddings from AlphaEarth, OlmoEarth-Nano, and Tessera, distributional pooling reduces geographic generalization gaps on high- N/D chips, but PASTIS acts as the failure case where covariance drops by more than 30 pp and first-order summaries such as *mean+std* and *center-weighted* remain competitive. We give practitioners a one-step decision rule and release *EuroSAT-Embed*, *PASTIS-Embed*, and benchmark code.¹

Index Terms—Geospatial foundation models, embeddings, pooling, land cover classification, crop classification, spatial generalization.

I. INTRODUCTION

A typical agricultural parcel in France covers 1.96 ha—196 pixels at 10 m resolution. A standard EuroSAT land-cover patch, by contrast, contains 4,096 pixels. Both settings require aggregating pixel embeddings from a geospatial foundation model (GFM) into a single patch-level vector, yet the number of pixels N available for that summary differs by a factor of twenty. We show this difference, relative to the embedding dimension D , determines which pooling method wins.

GFMs now publish dense pixel-level embeddings as downloadable data products [1]–[5], enabling practitioners to reuse them without access to the underlying encoder. Many downstream tasks—crop mapping [6]–[8], building detection, land-cover classification—assign a single label to a region rather than a pixel, requiring a post-hoc aggregation step (Fig. 1). This *input-label resolution mismatch* [9] is ubiquitous: EuroSAT [10] uses 64×64 patches, PASTIS [11] uses irregular agricultural parcels. The default aggregation—mean pooling—collapses each region to its average vector, discarding within-region variability that we show carries class signal.

Pooling is well-studied in image retrieval—generalized mean (GeM) [12], VLAD [13], NetVLAD [14], Fisher Vec-

tors [15]—and in temporal audio modeling [16], but remains largely unexplored for GFM embeddings over irregular-polygon regions. The key challenge specific to this setting is that N varies by three orders of magnitude across region types and sizes. When $N < D$, the sample covariance matrix has fewer data points than parameters and is mathematically rank-deficient; even at $N \approx D$ its eigenvalues are unstable, so second-order statistics estimated from such regions are unreliable. When $N \gg D$, those same statistics are well-conditioned and outperform mean pooling by a large margin. No prior work characterizes this transition or gives practitioners a decision rule.

We study post-hoc pooling across the full N/D spectrum. Using pixel embeddings from three GFMs on two benchmarks, we benchmark 11 training-free and 2 fitted pooling operators and measure how method rankings shift with region size. Our contributions:

- A practitioner-facing decision rule (Table III, Sec. V): the ratio N/D predicts when second-order operators (covariance, multi-moment stats) are viable, explaining the collapse of covariance pooling between EuroSAT ($N/D=64$) and small PASTIS parcels ($N/D \lesssim 1$).
- A benchmark of 11 training-free and 2 fitted operators on EuroSAT (3 GFMs, random and spatial splits, kNN and linear probes) and PASTIS, showing distributional statistics shrink the EuroSAT random→spatial accuracy gap from 9.6 pp to 3.4 pp on average (per-encoder reductions 43–76%) (Sec. V, Tables I–II).
- *EuroSAT-Embed* and *PASTIS-Embed*: pre-computed pixel-level embeddings from three GFMs, released with all benchmark code, providing the first controlled study of post-hoc pooling for GFM embedding products (Sec. II).

II. DATA AND EMBEDDINGS

EuroSAT. We use the 10-class EuroSAT land-cover dataset [10], which contains 27,000 Sentinel-2 patches at 64×64 pixels. We evaluate on both the standard random split [17] and a spatial split [18], [19] that partitions samples by longitude to reduce train-test leakage via spatial autocorrelation.

PASTIS. We use the PASTIS panoptic segmentation dataset [11], which contains 2,433 Sentinel-2 time-series patches covering France with instance-level annotations of ~85,000 agricultural parcels across 18 crop classes. Parcels are irregular polygons extracted from instance masks; parcel size varies from 5 to 3,440 pixels (median 196). We follow

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¹Code: GitHub; data: EuroSAT-Embed and PASTIS-Embed.

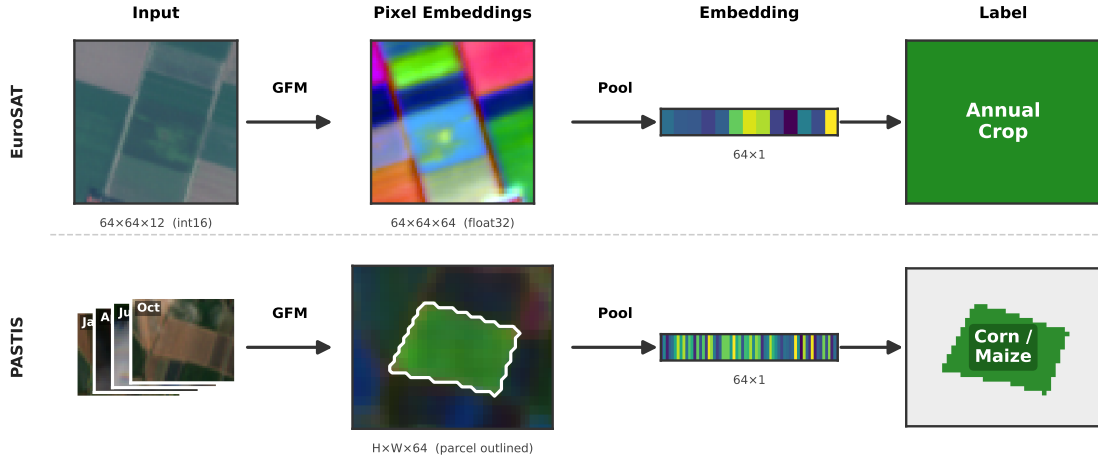


Fig. 1. **Embedding pipeline for EuroSAT (top) and PASTIS (bottom).** In both settings, a GFM emits one embedding vector per output pixel and a pooling operator aggregates those pixel vectors into one patch or parcel representation. *Top:* AEF and OlmoEarth consume a time series of image patches and return a patch of pixel-aligned embeddings; all 4,096 output vectors are pooled into a single patch vector for land-cover classification. *Bottom:* Tessera consumes a time series for each pixel and returns a pixel embedding; only pixels inside the parcel boundary (contour shown on a PCA pseudo-RGB visualization—first three principal components of the embedding rendered as colour) are pooled into a single parcel vector for crop-type classification. The number of pixels per parcel N varies from 5 to 3,440, making the ratio N/D a key design parameter.

the official PASTIS train/test split using the folds released with the original paper (folds 1–4 train, fold 5 test). Parcels with fewer than 5 pixels are excluded.

EuroSAT-Embed. We construct pixel-level embedding datasets from AlphaEarth [20] (AEF, 64-d int8), OlmoEarth-Nano [21] (128-d float32, $T=12$ subsampled time steps), and Tessera [22] (128-d float32). All three products ultimately provide one embedding vector per output pixel. AEF and OlmoEarth take a time series of image patches and return a patch of pixel-aligned embeddings, so each output vector can reflect neighboring-pixel context; Tessera takes a time series for each pixel and returns a per-pixel embedding without adjacent-pixel context. We apply the same three encoders to PASTIS parcels to produce *PASTIS-Embed*. We study pooling behavior, not encoder comparison.

III. POOLING METHODS

We benchmark training-free operators that compress an $H \times W \times D$ embedding tensor ($N=HW$ pixels) into a single feature vector, plus two operators that fit a light additional stage on the training split.

First-order statistics (D -d): *mean*, *max*, generalized mean (*GeM*) [12], and *center-weighted mean* (down-weighting boundary pixels).

Distributional statistics ($2D$ – $5D$ -d): *std*, *mean+std*, *mean+max*, *stats* (min/max/mean/std, $4D$ -d), *percentiles* ($5D$ -d), *median+IQR*, and *covariance* [23] (upper triangle, $D(D+1)/2$ -d).

Parametric (fit on train split): *PCA* (mean-pooled then projected) and *Bag of Visual Words (BoVW)* [24] (mini-batch k -means histogram).

In total we evaluate 11 training-free and 2 fitted operators.

IV. EXPERIMENTAL SETUP

We evaluate two probes: kNN ($k=5$, cosine distance) and multinomial logistic regression with regularization C selected

by 3-fold CV on the training split. We standardize before linear probing [25]. Both probes are deterministic under our protocol; rerunning the linear probe with three seeds (0, 1, 42) yields identical accuracy ($\sigma=0$ pp). EuroSAT results appear in Table I under random and spatial splits; PASTIS results (F1-macro) appear in Table II (OE = OlmoEarth-Nano, Tess = Tessera).

V. RESULTS

The N/D ratio governs second-order viability. The ratio of pixels per region N to embedding dimension D is a proxy for the conditioning of the sample second-moment estimator: the sample covariance is mathematically rank-deficient when $N < D$ and has unstable eigenvalues until N is several times D . This directly explains the collapse of covariance pooling between EuroSAT and small PASTIS parcels; the ranking of first-order operators (mean variants) shifts more smoothly with region size and is not predicted by N/D alone. EuroSAT patches have $N=4,096$ and $D \in \{64, 128\}$, giving $N/D \geq 32$: second-order statistics are well-conditioned. PASTIS parcels have a median $N=196$; for OlmoEarth and Tessera ($D=128$) the median sits at $N/D \approx 1.5$, and the smallest parcels in the dataset fall below $N=D$ entirely. Fig. 3 visualizes the transition; this single structural quantity explains every major finding below.

When $N \gg D$: distributional statistics win. On EuroSAT spatial splits, *stats* pooling reaches 91–95% accuracy (avg. 93.3%) vs. 81–92% for *mean* (avg. 86.3%), a +7.0 pp gain (Table I). *Covariance* leads on random splits for two of three encoders (OlmoEarth: 97.2%, Tessera: 97.6%). Fig. 2 shows that distributional methods are simultaneously more accurate and more robust: averaged across encoders, *mean* drops 9.6 pp from random to spatial splits while *stats* drops only 3.4 pp—a 65% reduction (per-encoder: AEF 63%, OlmoEarth 43%, Tessera 76%). *Mean+max* (4.0 pp) and *covariance* (4.8 pp) also exhibit small gaps. kNN trends are consistent, with *stats*

TABLE I

EUROSAT RESULTS (ACCURACY, %): LINEAR AND KNN ($k=5$) PROBES UNDER RANDOM AND SPATIAL SPLITS. BEST PER COLUMN APPEARS IN **BOLD**; SECOND-BEST IS ITALICIZED. DIM = OUTPUT EMBEDDING SIZE RELATIVE TO ENCODER DIMENSION D . **STATS** HIGHLIGHTED AS THE RECOMMENDED METHOD IN THE $N \gg D$ REGIME. *OLMOEARTH-NANO; OE = OLMOEARTH; TESS = TESSERA.

Pooling	Dim	Linear probe								kNN probe							
		Spatial split				Random split				Spatial split				Random split			
		AEF	OE*	Tess	Avg	AEF	OE*	Tess	Avg	AEF	OE*	Tess	Avg	AEF	OE*	Tess	Avg
Mean	D	81.1	91.8	86.0	86.3	95.3	96.4	96.1	95.9	85.5	87.8	84.7	86.0	94.6	94.8	94.9	94.8
Std	D	77.1	90.5	85.4	84.3	90.9	94.8	95.3	93.7	78.1	81.7	84.0	81.3	93.7	93.3	94.1	93.7
GeM	D	83.2	91.7	87.3	87.4	95.5	95.9	96.4	95.9	85.7	86.6	85.4	85.9	94.6	93.3	95.1	94.3
Center-Weighted	D	85.7	91.9	87.6	88.4	96.1	96.8	96.1	96.3	86.0	87.8	85.5	86.4	95.4	95.0	95.5	95.3
Max	D	86.1	90.6	89.8	88.8	92.9	92.9	93.7	93.2	80.5	85.5	80.8	82.3	90.5	91.4	92.5	91.5
Mean+Std	$2D$	84.2	93.4	94.8	90.8	96.1	96.6	97.1	96.6	86.4	88.3	86.7	87.1	94.9	94.9	95.3	95.0
Mean+Max	$2D$	89.1	93.3	94.4	92.3	96.1	96.3	96.5	96.3	85.3	87.9	86.9	86.7	94.7	94.4	95.8	94.9
Median+IQR	$2D$	81.7	93.2	88.4	87.7	94.8	95.9	95.9	95.5	86.0	86.0	85.7	85.9	94.1	94.0	94.4	94.1
Percentiles	$5D$	85.6	93.8	91.2	90.2	95.9	97.0	97.0	96.6	86.7	87.5	86.2	86.8	94.6	94.9	95.4	95.0
Stats	$4D$	91.0	94.0	94.9	93.3	96.2	96.6	97.3	96.7	86.9	87.6	89.3	87.9	95.0	94.4	95.6	95.0
Covariance	$D(D+1)/2$	89.1	93.4	93.7	92.1	95.8	97.2	97.6	96.8	84.4	84.8	89.8	86.4	93.4	94.3	94.5	94.1
<i>Parametric (fit on train split)</i>																	
PCA	64	79.7	91.6	80.8	84.0	95.2	95.6	95.0	95.3	85.3	88.4	86.1	86.6	93.5	94.1	92.7	93.5
BoVW	128	88.9	88.2	88.7	88.6	94.4	94.5	96.3	95.1	84.0	82.1	87.4	84.5	90.7	90.3	90.2	90.4

TABLE II

PASTIS RESULTS (F1-MACRO, %): LINEAR AND KNN ($k=5$) PROBES PER ENCODER. BEST PER COLUMN APPEARS IN **BOLD**; SECOND-BEST IS ITALICIZED. **MEAN+STD** HIGHLIGHTED AS THE RECOMMENDED METHOD FOR VARIABLE-SIZE REGIONS. OE = OLMOEARTH-NANO; TESS = TESSERA.

Pooling	Linear probe				kNN probe			
	AEF	OE	Tess	Avg	AEF	OE	Tess	Avg
Mean	64.5	54.7	49.0	56.1	56.5	41.1	43.8	47.1
Std	49.0	22.8	40.3	37.4	48.2	14.4	37.2	33.3
GeM	62.8	49.7	48.1	53.5	54.2	36.6	43.3	44.7
Center-Weighted	65.3	54.8	48.9	56.3	57.1	40.3	43.7	47.0
Max	57.4	38.8	45.5	47.2	46.0	27.6	39.3	37.6
Mean+Std	65.5	54.9	48.7	56.4	56.7	39.7	43.5	46.6
Mean+Max	65.0	54.4	48.8	56.1	53.0	35.2	42.7	43.6
Median+IQR	64.3	50.8	48.4	54.5	55.7	35.7	43.3	44.9
Percentiles	65.3	52.4	49.4	55.7	56.6	39.1	43.0	46.3
Stats	65.0	54.0	48.4	55.8	52.1	34.3	42.5	43.0
Covariance	58.7	42.4	47.0	49.4	47.6	22.4	39.6	36.6
<i>Parametric (fit on train split)</i>								
PCA	64.4	47.9	48.2	53.5	56.2	38.8	43.3	46.1
BoVW	44.7	23.0	15.5	27.8	43.6	22.1	14.1	26.6

leading on spatial splits (avg. 87.9%) and *center-weighted mean* on random splits (avg. 95.3%).

When $N \approx D$: PASTIS exposes the failure case. EuroSAT makes covariance pooling look attractive because every labeled patch supplies thousands of pixels. PASTIS tests the opposite regime: labels are irregular parcels, many parcels have $N < D$, and the sample covariance matrix becomes rank-deficient. In this setting, *covariance* falls from 89.1% spatial accuracy on EuroSAT-AEF to 58.7% F1 on PASTIS-AEF, and *stats* similarly loses its EuroSAT advantage (Table II). The practical lesson is not that PASTIS produces a large new accuracy win; rather, it shows where EuroSAT-derived pooling advice breaks. First-order summaries such as *mean+std* (linear avg. 56.4%), *center-weighted mean* (56.3%), and *mean* (56.1%)

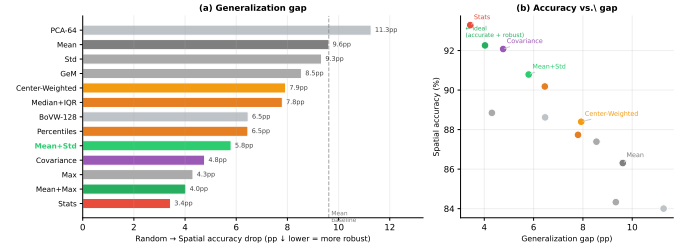


Fig. 2. **Distributional pooling is more accurate and more robust under spatial shift.** (a) Accuracy drop when moving from a random to a geographically disjoint split, averaged across three GFMs—shorter bars are better. Mean pooling drops 9.6pp; stats drops only 3.4pp, a 65% reduction. (b) Spatial accuracy vs. generalization gap per method. Methods in the upper-left are both accurate and robust; distributional methods cluster there while mean pooling sits lower-right.

are numerically close and remain competitive because they preserve coarse distributional structure without requiring stable second moments. For kNN, *mean* and *center-weighted mean* are the strongest PASTIS choices, so our variable-region recommendation should be read as a conservative first-order family rather than a single universally dominant operator.

Region-size breakdown. Fig. 4 ranks methods within four parcel-size bins averaged across all three embedding sources. For tiny parcels ($N < 50$), *center-weighted mean* leads (avg. rank 2.0), followed by *mean+max* (3.0) and *stats* (4.0). For small parcels (50–150 px) *center-weighted mean* again ranks first (2.0). For medium (150–500 px) parcels *mean+std* ranks first (2.67); for large (> 500 px) parcels *mean+max* ranks first (1.67), with *mean+std* third (3.67). *Covariance* sits near the bottom for tiny and small bins (avg. rank > 11) and recovers only to mid-pack (rank 8) in the large bin—it never leads at the parcel scale, consistent with N/D remaining $\lesssim 1$ –5 even for the largest PASTIS parcels at $D=128$. *Mean+std* is the only training-free method that stays in the top-3 for the small, medium, and large bins, making it the practical default for

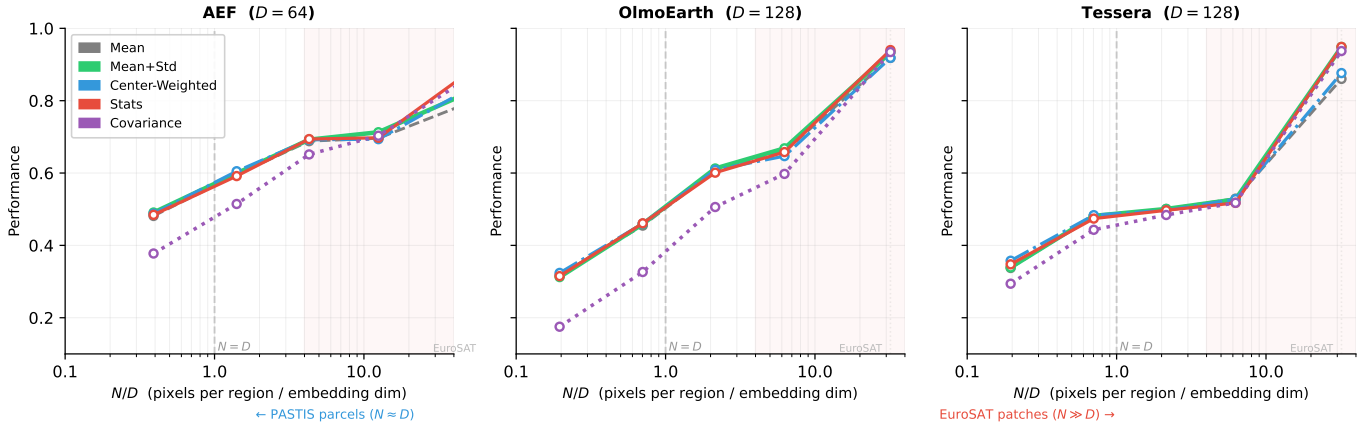


Fig. 3. **Pooling performance as a function of N/D , the ratio of pixels per region to embedding dimension.** Each panel shows per-bin F1 (PASTIS, four parcel-size bins) and accuracy (EuroSAT) plotted against N/D on a log scale for one GFM. The dashed vertical line marks $N=D$. To the left, where parcels are small and $N \approx D$, covariance (purple) collapses because its sample estimate becomes rank-deficient, while mean+std remains strong. Moving right toward EuroSAT patches where $N \gg D$, covariance recovers and leads. Mean+std stays competitive across the full range.

linear probes on variable-size regions; for similarity search, *mean* and *center-weighted mean* remain competitive defaults.

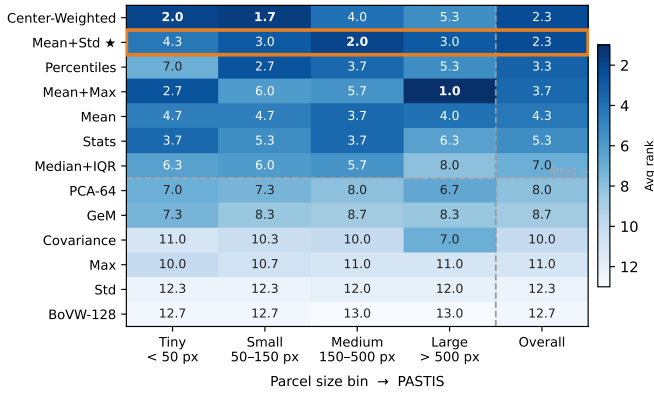


Fig. 4. **Method rankings shift as parcel size grows.** Each cell shows the average rank across AEF, OlmoEarth, and Tessera within each parcel-size bin—darker blue is a better rank (1=best). Center-weighted mean and mean+max dominate the tiny bin; mean+std rises as parcels grow; covariance moves from near-bottom (tiny/small) to mid-pack (large) but never leads at the parcel scale, since N/D remains modest. Mean+std (★, highlighted in orange) is the only training-free method that stays in the top 3 for the small, medium, and large bins. The dashed vertical line separates per-bin results from the overall ranking.

VI. DISCUSSION

Recommendation. Table III gives an empirical decision rule indexed by N/D : N/D formally predicts when second-order pooling is viable, and the small/large region rows are empirical defaults supported by our two benchmarks. When $N \gg D$ (large fixed-size patches, e.g. 64×64 at 10 m), use *stats* or *covariance* for peak accuracy; *stats* is more robust to distribution shift. When $N \approx D$ or regions are variable-size (parcels, fields, buildings), use *mean+std*: it wins the PASTIS linear-probe average (56.4% F1) and is the only training-free method that stays in the top-3 across the small, medium, and large parcel-size bins. For predominantly tiny regions ($N < D$), *center-weighted mean* dominates. *Mean* remains a useful lower

TABLE III

PRACTITIONER CHEAT-SHEET: PICK A POOLING OPERATOR FROM N (PIXELS PER REGION) AND D (EMBEDDING DIMENSION). APPLY AT INFERENCE TIME, NO TRAINING.

Regime	Recommended	Fallback
$N \gg D$ (fixed chips, $\gtrsim 1k$ px)	<i>stats</i> (or <i>covariance</i>)	<i>mean+std</i>
$N \approx D$ or variable-size regions	<i>mean+std</i>	<i>center-weighted</i>
Tiny ($N < D$)	<i>center-weighted</i> or <i>mean+max</i>	<i>mean+std</i>

bound. Accuracy and robustness move together here: *stats* and *mean+std* are simultaneously more accurate and more robust than *mean*, with no observed accuracy–robustness tradeoff.

Why distributional statistics help. Mean pooling collapses each region to its average vector, discarding intra-region heterogeneity. A residential neighborhood and an industrial zone may share a mean embedding while differing in variance: the mean erases that distinction; *stats* and *mean+std* preserve it. This signal is reliable only when N is large enough to estimate higher moments—which is precisely what N/D measures.

Practical applications. The decision rule maps directly onto GIS zonal-statistics workflows [26], [27], where raster values are aggregated over vector polygons (agricultural fields, building footprints, administrative units, ecological zones). As GFM embedding products become available at continental scale [1], practitioners running polygon-level inference—crop monitoring [6], habitat classification, urban mapping—can apply Table III at inference time with no additional training.

Related work. GFM embedding products have proliferated [2]–[4], [20]–[22], but post-hoc pooling for these embeddings is largely unstudied; the closest prior comparisons are in image retrieval (GeM [12], VLAD [13], NetVLAD [14], Fisher vectors [15]), bilinear/covariance pooling [23], and aggregation in temporal audio [16]. Our setting differs in that N varies by three orders of magnitude across regions, which is what makes the N/D ratio actionable.

Limitations. We evaluate on EuroSAT (10 classes) and PASTIS (18 crop classes), both Sentinel-2 at 10 m, with three embedding sources; broader coverage (e.g., BigEarthNet [28], fMoW [29], multi-label or dense-prediction settings) would strengthen conclusions. OlmoEarth is evaluated with $T=12$ subsampled time steps; AEF and Tessera use the encoders' default annual representation. The temporal aggregation strategy per encoder is fixed rather than ablated. We study only patch- and parcel-level labels, not semantic segmentation. We focus on training-free pooling; learned aggregation [30] may improve further but requires encoder access.

VII. CONCLUSION

The pixel-to-dimension ratio N/D is a principled, single-number criterion for whether second-order pooling of GFM embeddings is viable, anchoring an otherwise empirical operator choice. We give a one-step recommendation in Table III that practitioners can apply at inference time, and release *EuroSAT-Embed*, *PASTIS-Embed*, and benchmark code to reproduce every result.

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